AI for Testers

V2.3.2

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Who We Are

Coveros helps organizations build better software faster.

We provide consulting, coaching, and learning opportunities to enterprises, teams, and individuals.

Coveros is a GitHub Premier and Microsoft Partner.

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Our Services

Software Process Improvement

Services to help you modernize your software process by improving agility, continuous delivery, test automation, and application security leveraging AI.

Software Development

Services to support the planning, design, development, testing, security, and delivery of mission-critical software applications leveraging AI.

Immersive Learning

50+ hands-on training classes taught publicly and privately.

Enterprise, team, and individual coaching to reinforce concepts.

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Community Support

Software Conferences

STAR EAST
A TECHWELL EVENT

April 26 – May 1, 2026
Orlando, FL + Online

AI Con USA
A TECHWELL EVENT

June 7-12, 2026
Seattle, WA + Online

STAR WEST
A TECHWELL EVENT

September 20-25, 2026
Anaheim, CA + Online

Coveros customers receive discounted or free conference passes based on volume of business

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Who We Help

- Retail
- Healthcare & Devices
- Finance
- Technology & Security
- Communications

30,000+
Software professionals trained

200+
Organizations assessed and improved

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Administration

Course Timing

Breaks

Meals

Electronic Devices

Facilities

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Student Introductions

- Your name
- Your role in your organization
- Experience
  - Testing software
  - Automating test cases
  - Artificial intelligence
- Expectations
  - What do you wish to learn?

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Learning Objectives

At the conclusion of the course students will be able to:

- Understand how AI can assist the testing process
- Use AI to assist test planning and management
- Perform testing using AI
- Use AI to assist in automated test case generation
- Leverage AI for data generation and predictions

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Course Layout

- Topic 1: Introduction to AI-Assisted Testing
- Topic 2: AI-Assisted Test Planning
- Topic 3: AI for Test Data Management
- Topic 4: Exploratory Testing Using AI
- Topic 5: AI-Assisted Test Automation
- Topic 6: AI for Test Results Analysis and Reporting

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Environment Setup

Testing Access to the Training Environment

Windows 11

Open Windows Remote Desktop (RDC)

Remote Desktop Connection

Download and Install Windows App

IP Address
From instructor
Username
From instructor
Password
From instructor

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Introduction to AI-Assisted Testing

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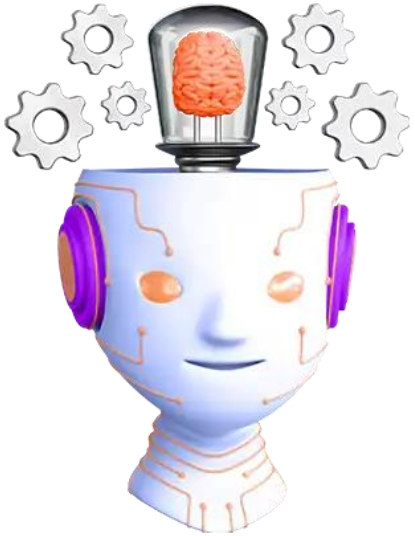
What is AI-Assisted Testing?



AI-Assisted Testing is the application of AI techniques to enhance the testing process and increase tester productivity

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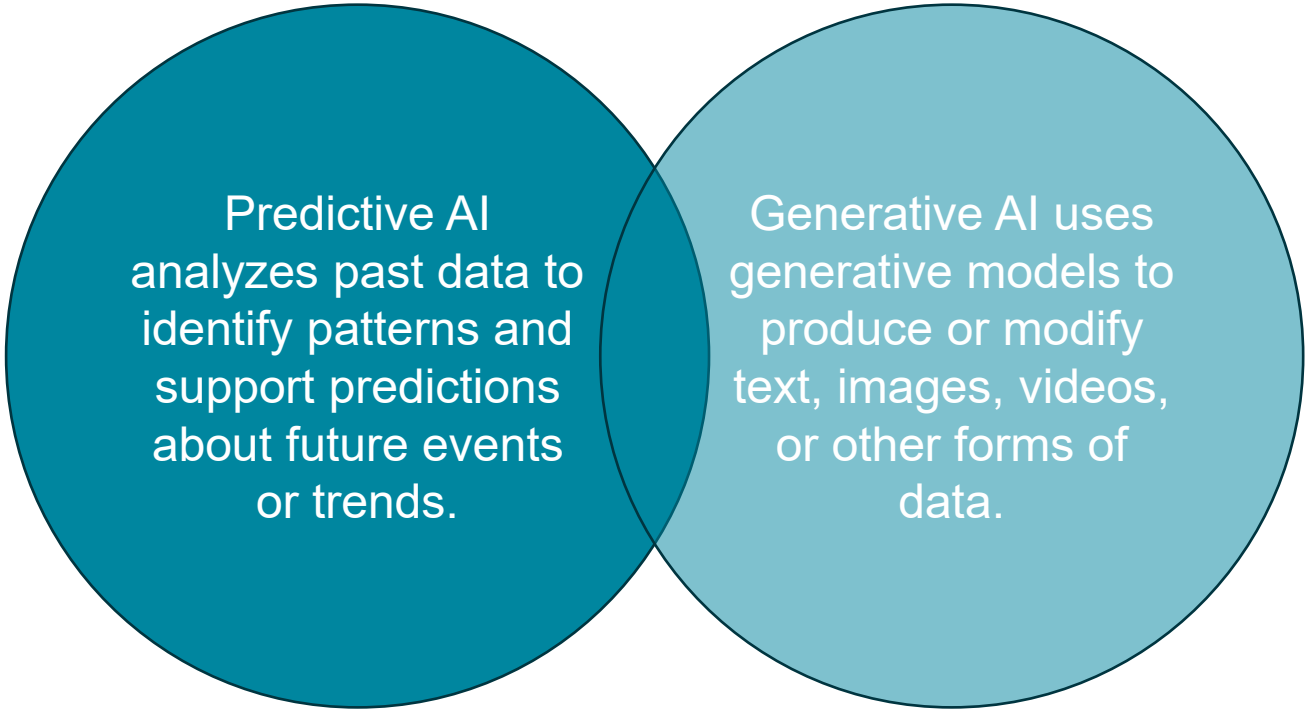
Predictive vs. Generative AI Techniques



Often based on machine learning (ML)

Often based on large language models (LLMs)

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What Generative AI Models Can Do

- Generate content (text, audio, images, video, code, tests, etc.)
- Enhance or augment existing content
- Transform content into another format
- Summarize content
- Make predictions and support decision-making using patterns found in data

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AI-Assisted Testing Activities

Risk-Based Testing Process

Analyze reqs & applications	Analyze defects and tests	Generate and transform test data	Generate test cases and test scripts	Transform tests and other content
AI Support for Testing	Identify risks and tests that mitigate risks	Document tests, scripts, test reports	Perform testing	Analyze test results

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Benefits in Using AI for Testing

- Increased efficiency by reducing toil and automating analysis
- Improved test coverage by identifying untested areas faster
- Reduced testing time by automating existing manual activities
- Provides a ‘pair tester’ to bounce ideas off and to provide additional suggestions

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Ethical Considerations and Challenges

- While AI can greatly benefit software testing, it does not replace software testers!

- Some of the ethical considerations and challenges testers face include:
  - AI model hallucinations
  - Intellectual property and data privacy issues
  - AI models trained on inaccurate, false, or inadequate data
  - Silver bullet syndrome
  - AI model bias and fairness

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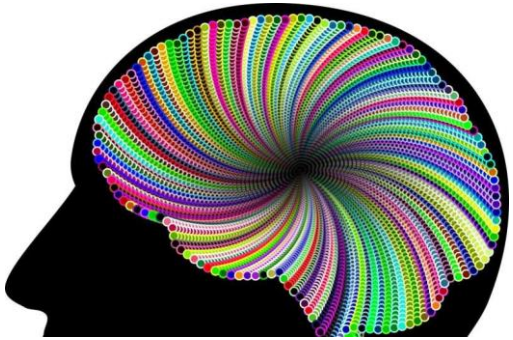
AI Model Hallucinations

Hallucinations



- Generative AI models sometimes give incorrect answers called hallucinations
- These answers may contain fabricated information, go against common logic, or be nonsensical
- Hallucinations happen due to inadequate model training, training on bad data, overfitting, or underfitting

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IP and Data Privacy Issues

IP and Data Privacy

- AI model licenses may not protect proprietary information (IP) fed into them
- Customer data used during model training and testing may violate privacy laws

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Quality of Training Data

Quality of Training Data

- ‘Garbage in garbage out’ applies to AI models
- Models may be overfit or underfit

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Silver Bullet Syndrome

Silver Bullet Syndrome

- It is tempting to blindly trust the results of a prediction or generated content that comes from an AI model
- Remember that AI cannot replace your context, reasoning skills, and critical thinking

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Bias and Fairness

Bias

- The fairness of an AI model depends on the data that was used to train it.
- For a particular purpose, an AI model might not meet the definition of fair
- Understand the AI model you use and ensure it fully represents your use cases.

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Prompt Engineering for Testers

Best Practices for Software Testing

- Principle #1: Write testable prompts
- Principle #2: Give models encouragement to think
- Principle #3: Build good prompts iteratively
- Principle #4: Think beyond content creation

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Prompt Principle #1

Write testable prompts

- Use delimiters to clearly describe what you want
- Define the structure of the response you expect
- Use examples to guide responses
- Describe your role for additional context
- Be focused and specific … don’t try to solve it all in one shot

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Prompt Principle #2

Give models encouragement to think

- Specify the steps to complete the task you have requested
- Instruct the model to work out its solution before responding and output its thinking along with its answer
- Offer the model an incentive to rethink or improve its answer
- Ask the model to compare multiple answers and choose the best one … considering using various models if available

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Prompt Principle #3

Build good prompts iteratively

- Use prompt iteration and few shot prompting to refine answers
- Understand the context window size for your LLM as that will constrain the knowledge a model has to answer you
- Remember that too many iterations will cause hallucinations as overfitting will occur

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Prompt Principle #4

Think beyond original content creation

- Use AI models to transform test data and test results into the format you need
- Summarize test activities and results
- Use AI models to update and improve test documentation
- Use AI models to review your work and suggest improvements

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Prompt Libraries

- As you craft prompts, you will identify ones that make sense to save and reuse
- Sometimes reuse will involve modification of prompts to make them more context-specific
- Creating a prompt library that maintains effective prompts for reuse should be done by every test team using AI
- We will use a prompt library in this class to support the exercises and give you a starting library to take home and use

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What is Visual Studio Code?

- VSCode is a lightweight, powerful integrated development environment (IDE) supporting hundreds of programming languages
- Customizable with an extensive library of extensions you can plug in
- Integrated debugging and version control
- Offers syntax highlighting, code completion, and automatic indentation out of the box

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Case Study

Introduction

- A software startup is creating a wine classification application
- Based on various chemical properties, the software will determine which variety of wine from Italy it is
- This startup has no dedicated software testers, so they have hired you to test this product

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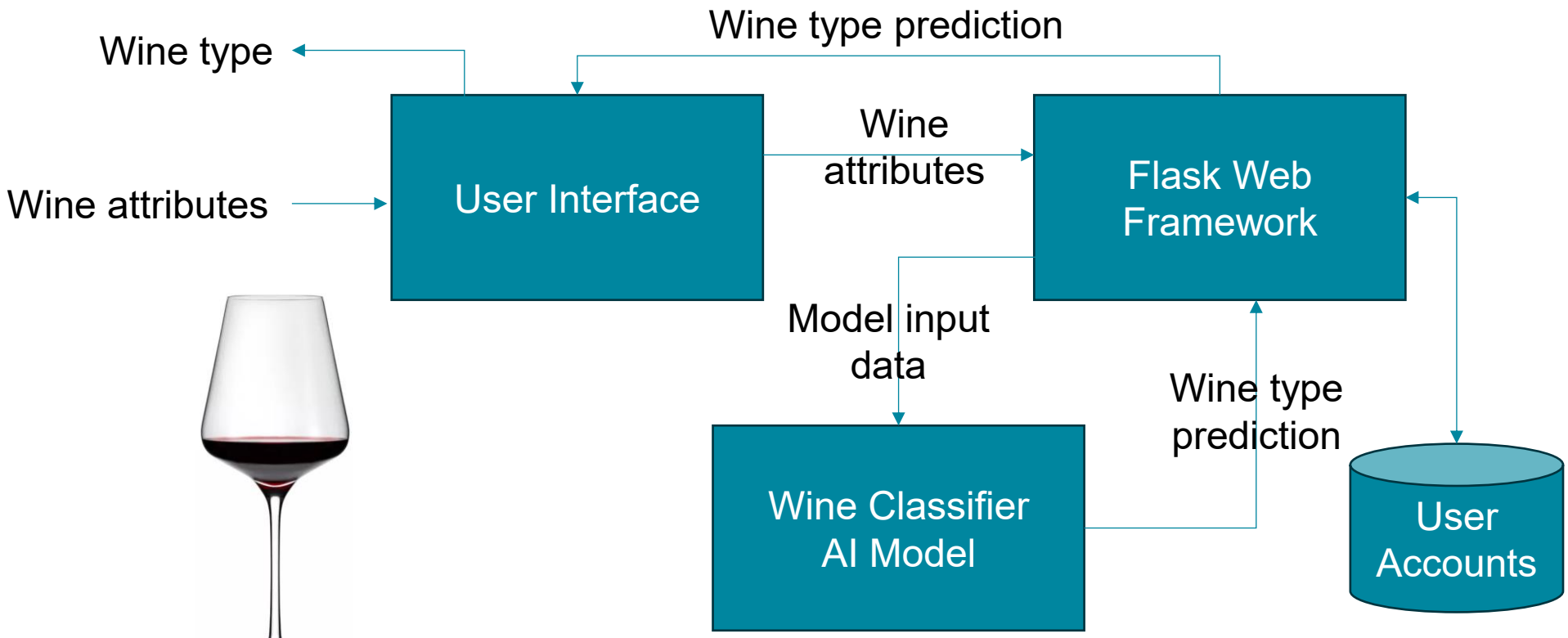
Wine Classification Application

Product Architecture



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Workshop Demo and Exercise #1

Explore wine classifier application

Wine classifier

- Follow along as the instructor downloads the wine classifier application and starts it up
- Watch as the instructor runs the wine classifier application (wine.py)
- Break into teams and
  1. Create a team name
  2. Run the wine classifier application
  3. Play around with its features
  4. Decide who will test which feature
     - Logins, Registrations, Profiles, Classifications

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AI-Assisted Test Planning

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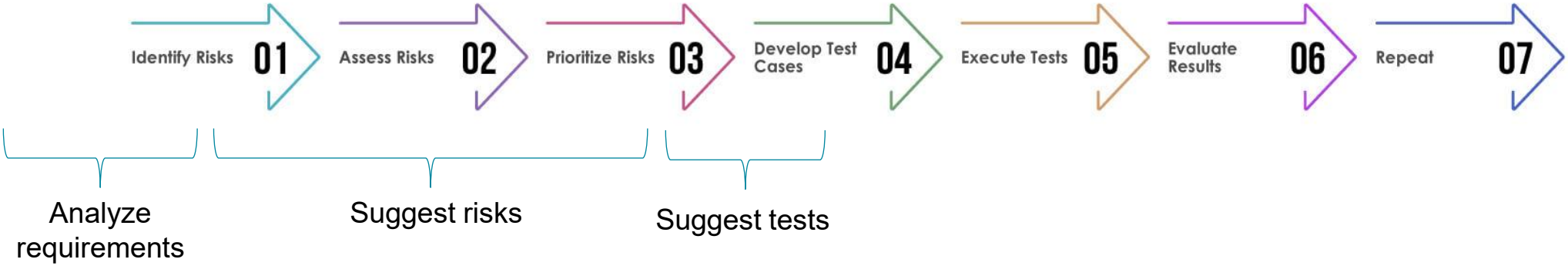
AI-Assisted Test Planning

Risk-based testing process (revisited)



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AI-Assisted Test Planning and Management

AI Tool Support

Generative AI Chatbots

Gemini

ChatGPT

COPILOT

AI-based Test Management Tools

spiraTeam

TestLink

QMETRY
A SMARTBEAR COMPANY

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Using AI to Analyze Requirements

- Generative AI models are good at analyzing existing documents.
  - Works for traditional requirements formats
  - Works for user story / use case requirements formats

- If you have high-level business requirements associated with your functional/non-functional requirements, use them as context.

- If your requirements are not in the format, you want them, specify a format and ‘translate’ them as part of your requirements process.

- It is not recommended to try and generate requirements using generic descriptions of your application

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Using AI to Analyze Requirements

- Provide as much context as possible on project goals, user needs, timelines, previous versions of requirements, etc.

- If you ask an AI model to analyze all of your requirements at once the results will
  - Identify cross-cutting concerns and more holistic insights
  - Be more efficient for large projects than analyzing requirements one at a time
  - May struggle to provide enough specific advice due to the size of the task

- If you ask an AI model to analyze requirements one by one the results will
  - Be more focused with results that are easier to rework/debug
  - Be more effective for smaller projects as the results will be more specific and potentially useful
  - Will not give cross-cutting advice

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Using AI to Analyze Requirements (cont.)

- Consider a hybrid approach to get the best of both worlds
  - Hybrid approach 1: Analyze key subsets or groups of related requirements together.
  - Hybrid approach 2: Analyze individual requirements initially and then use the results to review the entire set for consistency and completeness.
- Use experimentation on a small set of requirements to determine which approach makes the most sense for your specific situation

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Workshop Demo

Improve Requirements with AI

- Follow along as the instructor shows how to analyze business requirements for our wine classification example
  - business_requirements.ipynb
  - business_requirements_generic_prompt.ipynb
  - business_requirements_specific_prompt.ipynb
- Notice the differences between a generic and specific prompt
- Performing a hybrid prompt is left for the exercises

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Using AI to Suggest and Improve Risks

How AI can help test planning and management

- The same alternative approaches that we considered for requirements analysis apply to identifying risks
  - Analyzing requirements one by one for risk identification
  - Analyzing all requirements at once for risk identification
  - Hybrid approaches
- Including information on the types of risks you are most concerned about in your prompts … these ideally have been identified by your business stakeholders
- If you maintain a risk register or other risk documentation, consider including information on what you want the resulting risk information to look like

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Using Models to Improve Risk Prompts

How AI can help test planning and management

- Models represent views of your system
  - Functional decomposition
  - Dataflow
  - Heuristic
  - Use case or workflow

- Understanding various representations for your system can help focus your prompts and get better AI results

- Use this to specify better prompts for identifying risks and resulting test cases

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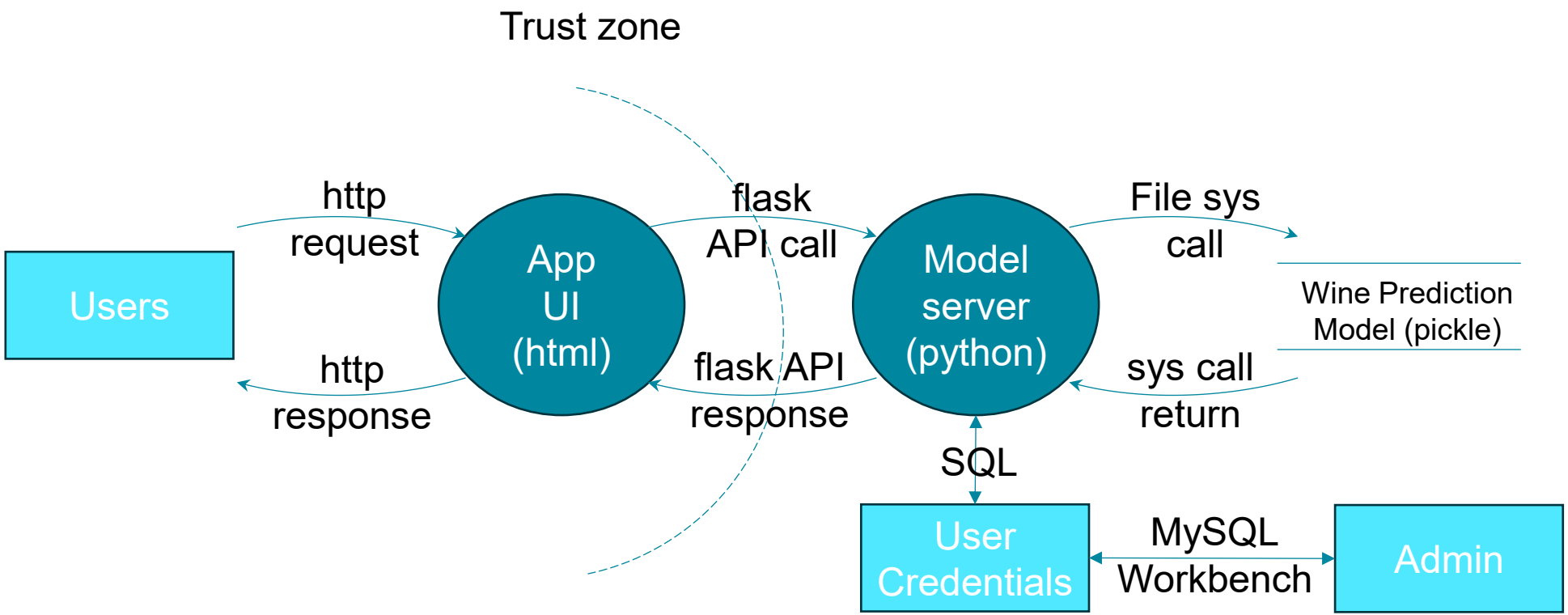
Using Models to Improve Prompts

A dataflow model of our wine classification system



What information in this model may help us prompt better?

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Workshop Demo

Using AI to Suggest and Improve Risks

- Follow along as the instructor shows how to generate and improve risks from various types or requirements
  - logout_epic_risks.ipynb
  - logout_stories_risks.ipynb
  - logout_model_risks.ipynb
- Notice the difference in results for these various levels of prompts

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Using AI to Generate Tests

- Tests are often broken down into several levels during test planning
  - Test requirements – the high-level description of a particular test
  - Test specifications – the detailed steps associated with a test requirement
- In agile, tests aligned with user story acceptance criteria plus automated test steps replace formal test documentation
- We will be using an agile testing approach in this class

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Using AI to Generate Tests

- Leverage the results of your requirements analysis, your risk analysis, and any existing test cases you have created when prompting to create tests
  - Be explicit that you are seeking test cases that do not overlap with your existing ones
- Send in context for tests to improve results
  - What level of test cases do you want?
  - What testing techniques do you want to use?
  - In what format do you want the resulting tests?
- Use results from risk analysis to identify key negative tests to perform

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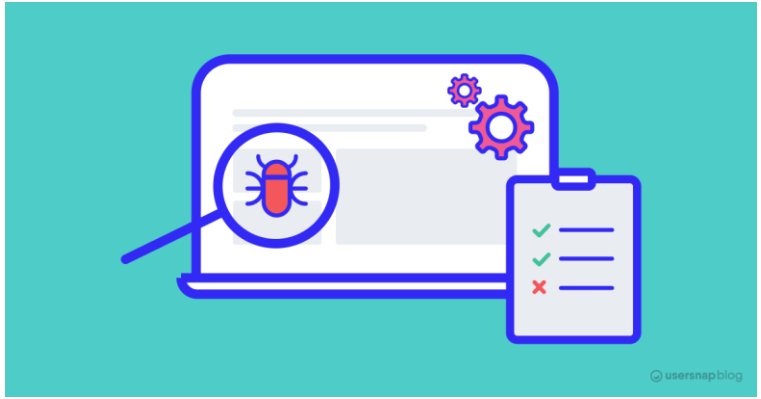
Workshop Demo

Using AI to Generate Tests



- Follow along as the instructor shows how to generate good tests from a set of user stories
  - logout_test_results.ipynb
- Creating tests leveraging risks and model information is left for the exercises
- Our prompt library includes examples you can use when doing this during exercises

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Workshop Exercise #2

Test Plan with Generative AI

- Break into our teams
- Use AI to perform test planning on the wine classification user stories you are responsible for
  - Login User Stories: login_user_stories.ipynb
  - Register User Stories: register_user_stories.ipynb
  - Profile User Stories: profile_user_stories.ipynb
  - Classification User Stories: classify_user_stories.ipynb
- For stories in each feature do the following steps:
  - Analyze user stories to understand and improve them
  - Generate risks for a few important quality characteristics
  - Create positive and negative tests
- Leverage the existing prompt library to help you get started!

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Commercial AI-Based Test Management

- SpiraTeam is a commercial test management solution
- It provides end-to-end traceability for risk and compliance
- Works for agile and traditional software processes
- Incorporates AI into the process to automate the generation of risks, user stories, and test cases

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Workshop Demo

Improve Requirements with AI

- Follow along as instructor shows how to use products like SpiraTeam to
  - Identify risks
  - Create tests
- Discuss the pros and cons of using these tools vs. chatbots

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AI for Test Data Management

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Test Data Management

- Creating and managing test data is an essential part of software testing
- But as the complexity of our apps grows, so does the complexity of our test data
- Effort spent creating test data, repurposing production data for testing, augmenting test data, and masking/anonymizing test data is time intensive
- Traditionally test data generation has been done using:
  - Commercial test data management solutions (manual and automated creation/mgmt.)
  - Libraries like Python's Faker if you are writing your own custom test data generators
  - Manual creation of datasets as needed

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Aspects of Test Data Management AI Can Assist With

- Test Data Generation
- Data Masking and Anonymization
- Test Data Quality
- Test Data Transformation

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AI for Test Data Generation

- AI leveraging LLMs provides an approach for automatically generating test data from scratch
- If you have existing test data, you can also augment it to increase coverage or have more testing data available
- When augmenting existing test data that includes expected results, AI can generate expected results for new test data … but it won't always get this right

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AI When Generating Test Data from Scratch

- Give the chatbot a specification for the information you want to generate, the ranges of values for that information, and the data types
- This information can be in table form using delimiters to help guide it
- For complex relationships you may need to run your model twice
  - First time based on your specification for data fields
  - Second time based on dependencies between fields
- If your data changes over time you prompting to specify these changes

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AI for Test Data Augmentation

- Test data augmentation refers to creating additional test data leveraging existing test data.
  - Some call this synthetic test case generation
- Prompts incorporate existing test data for context and instruct a large language model to analyze it and generate appropriate test data.
- If inputs are visual, image transformation is often used
- While not foolproof, analyzing the expected results of existing test data can help provides expected results for the synthetic data as well.

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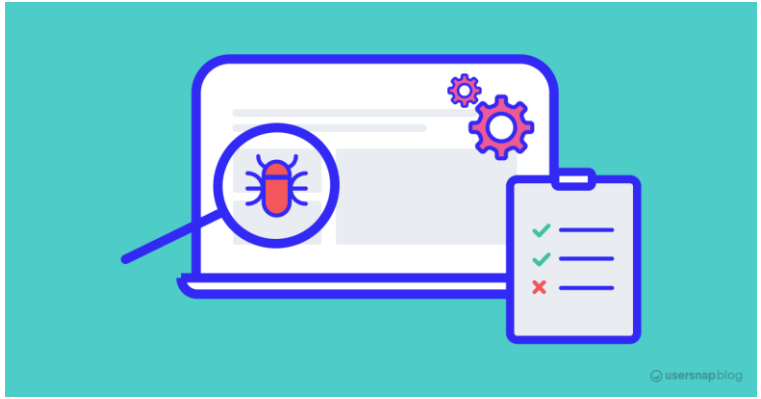
Workshop Demo

Using AI to Generate Test Data



- Follow along as the instructor shows how to generate test data
  - test_data_generator.ipynb
- Demonstration will include:
  - Generating test data from scratch
  - Generating additional test data
  - Generate negative test data

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AI to Mask and Anonymize Data

- Traditionally masking and anonymization of data has been done with libraries like Python's Pandas and other data analysis packages or commercial test data management products
- Approaches:
  - substitution,
  - redaction,
  - randomization,
  - generalization
- Large language models provide a different approach that is easier for non-coders to leverage
- Specify in your prompts the fields to modify and the approach to use

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AI to Mask and Anonymize Data (cont.)

- Generative AI models can also help decide the best approach to masking and anonymizing data
- Ask a model what approach makes sense for a particular scenario
- If there are correlations between any of the data you are masking or anonymizing, make sure you ask your model to assure these relationships are maintained

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AI to Improve Test Data Quality

- Traditionally data cleaning has been done with libraries like Python's Pandas and other data analysis packages to increase data quality
- Large language models provide a different approach that is easier for non-coders to leverage
- Prompting with guidance about how to address data quality issues can automatically correct these issues in existing test data

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AI to Improve Test Data Quality

- Typical test data quality and completeness issues
  - Missing values
  - Data inconsistencies
  - Duplicate data
  - Incorrect types
  - Data errors

- By asking a model to review your data set and identify these issues, you can quickly generate higher quality test data sets

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AI for Test Data Transformation

- Transforming data between data structures is simple if you give the chatbot enough information when prompting
- Tell it the existing format of your data
- Tell it the format or structure you want the data to be transformed into
- Your results can just be data or code that processes the data (e.g., SQL queries)

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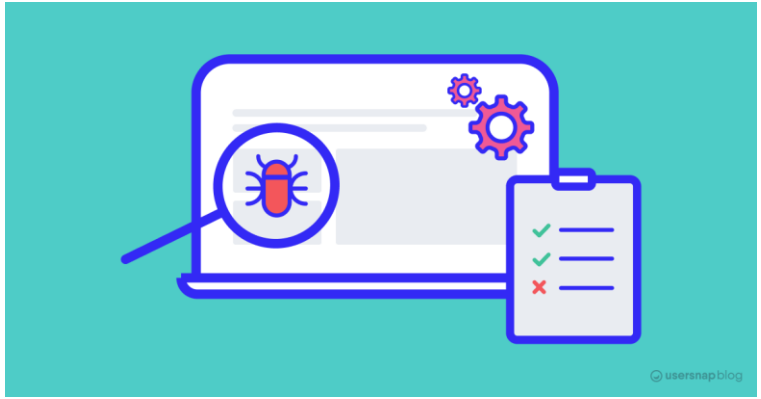
Workshop Demo

Using AI to Modify Test Data



- Follow along as the instructor shows how to modify test data
  - test_data_cleanup.ipynb
- Demonstration will include:
  - Anonymizing data
  - Improving data quality
  - Transforming data sets

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Integrating LLMs to Support Test Data Generation

- Basic code can allow you to incorporate AI models into your testing framework to dynamically generate, augment, clean, and transform your testing data
- Use API to LLM model to prompt
- Process data returned to use results

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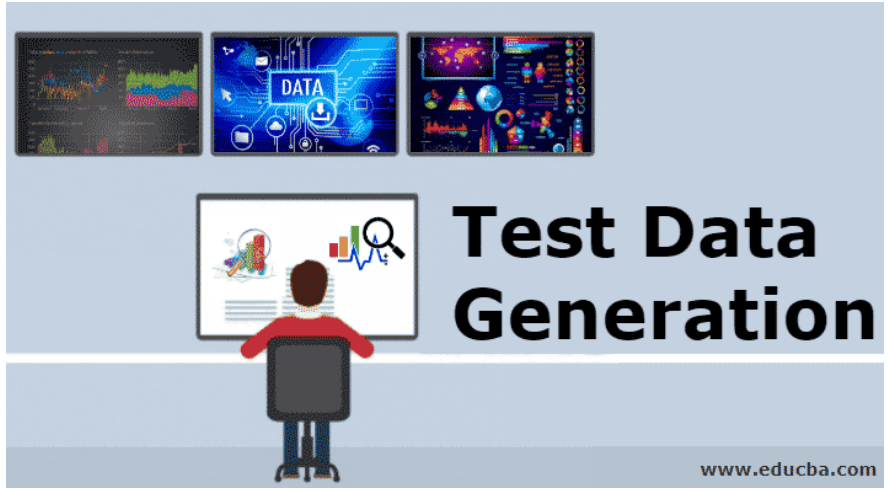
Workshop Demo

Call an AI model for your test framework



- Follow along as instructor shows how to access an AI model from test framework code
- What are the advantages/ disadvantages of calling AI models from your test code?

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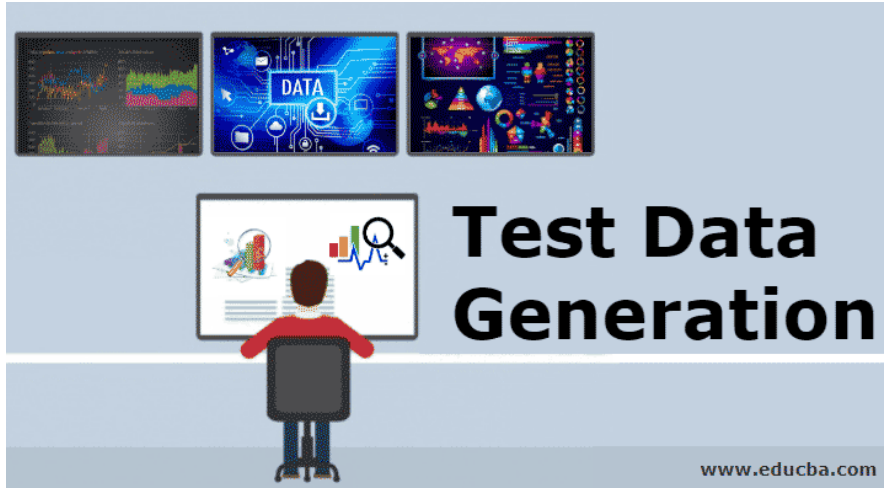
Workshop Exercise #3

Creating Test Data



- Create better test data for testing the wine characteristics wine feature.
- The existing dataset is in wine.csv
- Break into teams and work as a team to:
  - Identify and correct quality issues with the existing test data set
  - Double the size of the existing test data set
  - Convert the data set from CSV format to JSON format

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Exploratory Testing Using AI

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What is Exploratory Testing?

Exploratory testing is a software testing approach where testers simultaneously learn, design, and execute tests.

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Session-based Exploratory Testing

Steps in a session-based exploratory testing process:

- Create charters and determine who will focus on what
- Perform initial product exploration
- Observation and learning
- Design and execute new tests
- Continue learning and adapt
- Document findings

- Charters can take many forms but a common one[1] is:
  - Explore <target>
  - With <resource>
  - To discover <information>

[1]*From the book Explore It*

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Use AI to Create Test Charters

- Start with a User Story or Epic (or another form of a requirement) that includes acceptance criteria
- Identify initial risks in the requirement if you haven't already
- Prompt for additional risks for a particular quality attributes you want to focus on in this testing effort
- Translate risks into initial charters in the proper format
- Prompt to create additional charters based initial charters and existing risks

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Using AI During Exploratory Testing

Charter Approach

- Identify a test charter to pursue
- Perform exploratory testing against the charter to identify issues
- When you have run out of ideas, use a mnemonic to have AI generate additional ideas
  - SFDPOP – Structure, Function, Data, Platform, Operations, Performance
  - PAOLO – Portrait view, Audio, Objects, Landscape view, Overlays
  - CUBES – Characteristics, Use, Boundaries, Errors, Structure
  - Give AI information on how you want the results structured

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Using AI During Exploratory Testing

Charter Approach (cont.)

- Make sure you have taken notes (text, audio, images, event capture) to capture:
  - How you approached our exploratory testing sessions
  - What setup was required
  - What testing was done
  - What was learned from our testing
  - What bugs were discovered

- Consider using automation and AI to help capture information
  - Button/keystroke capture software
  - AI summarization capabilities
  - If being done virtually, consider recording your session as a Zoom meeting with AI assistant turned on

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Freestyle Exploratory Testing

- Break down your application by screens on your UI
- Take screenshots of related UI screens
- Prompt a chatbot to identify bugs, issues, or important things missing on each screen
- Give as much context as possible including what quality characteristics you are most interested in
- Make sure you have taken notes (text, audio, images, event capture) as discussed for Session-based Exploratory Testing

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Documenting Exploratory Testing Results

- While testing teams often do a debrief after exploratory testing sessions, there are advantages to formally documenting what was done:
  - Assures we don't repeat testing that doesn't need to be repeated
  - Provides downstream evidence and test results for compliance and audit
  - Provides AI data to analyze your exploratory testing process and defects
- Use prompts to translate notes into a full report
- Guide the structure of the report and keep it consistent
- Maintain these results in folders by charter, version, etc.

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Analyzing Your Exploratory Testing Results

- Use AI to review your charters and exploratory testing results to identify gaps in your testing process
- Use AI to review historical defect information contained in your exploratory testing documents to pinpoint areas of risk in your product
  - Feed this information back to the development team
- Use AI to review historical defect information contained in your exploratory testing documents to help improve your software development process
  - Feed this information back to the development team

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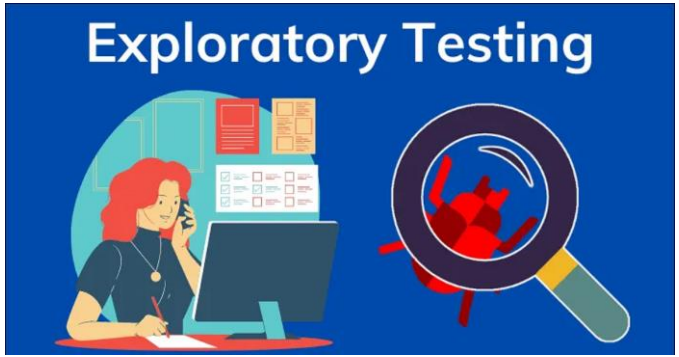
Workshop Demo and Exercise #4

Exploratory Testing Using AI



- Follow along as the instructor shows how to use AI for exploratory testing
  - exploratory_charters.ipynb
  - about_screen.png

- Break into teams
  - Create a charter for one of your user stories
  - Perform some exploratory testing
  - Feed a UI screen into a prompt and see what suggestions it has
    - classify_wine_screen.png
    - user_login_screen.png
    - user_profile_screen.png
    - user_reg_screen.png

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5

AI-Assisted Automated Testing

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What is AI-Assisted Automated Testing?

The process of leveraging Generative AI capabilities to assist in the automation of testing

GitHub Copilot

tabnine

codeium
Windsurf Editor

amazon
Amazon Q

CURSOR

codota

Continue

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AI-Assisted Automated Testing Capabilities

- *Code Completion* – using AI to help create test code
- *Automated Test Case Generation* – using AI to help generate automated tests
- *Test Case Debugging* – using AI to fix test execution problems
- *Refactoring Tests* – using AI to improve tests
- *Translation* – using AI to translate tests and test data to different formats
- *Test Documentation* – using AI to create and improve documentation

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AI-Assisted Code Completion

- Aspects of Code Completion
  - Suggests completion of line(s) as you type
  - Suggests completion of comments as you type
- Most AI Assistants use the Tab key to accept suggestions
- Many allow you to choose a suggestion from a list of alternatives
- Code Completion works for any type of code
  - Application, Test Scripts, Infrastructure as Code, etc.

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AI-Assisted Automated Test Case Generation

- There are a variety of ways to generate test cases with AI Assistants
  - Chat window
  - Inline chat
  - Directed comments
- You can accept or reject the code you generate
- Be specific in your prompts to get good results!
- Context and examples help a lot

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AI-Assisted Test Case Debugging

- When tests you run fail, AI Assistants can help you debug them
- By referencing the ‘terminal’ that ran the failed test, you can diagnose the issue
- The AI Assistant will offer suggestions for correcting the problem

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AI-Assisted Refactoring

- Refactoring is restructuring code to improve operation without altering functionality
- Often done to clean up ‘code smells’ or adjust code to meet coding standards
- Most AI Assistants provide a mechanism for highlighting code and refactoring it automatically
- You can accept or reject suggested refactors

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AI-Assisted Test, Test Data, Test Results Translation

- Test translation is about translating your tests from one language or framework to another
- Test data translation is about translating your test data from one format to another, e.g., CVS to JSON
- Test results translation is about translating results from one format to another

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AI-Assisted Automated Test Documentation

- AI-Assistants can generate documentation for your test code with commands such as /doc
- Code completion capability can be used to create comments as you create your tests
- Chat window can be used to comment selected test code
- Chat can be used to /explain test code and generate information that can be used as its specification

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Workshop Demo

AI-Based Automated Test Case Generation

Visual Studio Code

GitHub Copilot

- Intro to Coding Assistants
  - Code completion
  - Generate a new test
  - Refactor an existing test
  - Debug a failed test
  - Change testing framework
  - Comment a test
- Show how to run tests in VS Code

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Workshop Exercise #5

AI-Based Automated Test Case Generation

Visual Studio Code

GitHub Copilot

- Break into teams
- Create a new automated tests for one of your user stories in your epic
  - tests/test_classifier.py
  - tests/test_login.py
  - tests/test_register.py
  - tests/test_profile.py
- Try to run it and see if it works
- Debug it if it doesn't

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AI-Assisted UI Testing

- There are pros and cons to having AI generate UI-based tests from scratch

- Pros
  - Generates code that works (mostly) very quickly
  - You don’t need to remember syntax to get working tests
  - Can work well if you have existing tests to base new ones on

- Cons
  - Without context you can end up spending more time reworking a generated test than just creating it manually!

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AI-Assisted UI Testing

- As an alternative to auto generating complete tests, AI can support individual aspects of the test automation process
  - AI-based locator selection
  - Image recognition of UI features
  - Support when using page objects
  - Self-healing of tests

- Some of these features are ones that code assistants can help with while others use more traditional machine learning and computer vision algorithms

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Test Tools with AI Capabilities

functionize

mabl

applitools

AI powered Visual Testing & Monitoring

testRigor

Rapise

Playwright

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What is mabl?

- mabl is a commercial test automation solution
- Provides automation of tests at a variety of levels using a low-code approach
- Advanced AI capability to generate and update test automation scripts

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Workshop Demo

AI-Assisted UI Testing

mabl

- Watch as instructor demonstrates mabl automated test case generation
- Watch as instructor demonstrates self-healing of tests when things change

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6

AI for Test Analysis and Reporting

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Using AI for Test Analysis and Reporting

- Automate Test Report Generation and Review
- Test Documentation
- Enhance Defect Management
- Streamline regression testing
- Defect prediction

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Automated Test Report Generation

- Summarizing Test Results: AI can analyze test logs, screenshots, and other test artifacts to automatically generate concise and informative test reports.
- Creating Structured Reports: AI can structure test reports with clear sections for test objectives, test cases, test results, defects, and conclusions.
- Generating Visualizations: AI can create charts and graphs to visualize test results, such as pass/fail rates, defect trends, and test coverage.

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Test Case Documentation

- Generating Test Case Descriptions: AI can help generate clear and concise descriptions of test cases based on user stories, requirements, and other relevant information

- Automating Documentation Maintenance: Use AI to automatically update test case documentation as your tests change

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Enhance Defect Management

- Generate bug reports: AI can assist in creating detailed bug reports by automatically extracting relevant information from test logs, screenshots, and other sources
- Improve bug descriptions: AI can help improve the clarity and conciseness of bug descriptions by suggesting more precise language and identifying missing information
- Support defect triage: AI can analyze bug reports to identify bug categories, prioritize them for faster resolution, and assign them
- Support defect prediction

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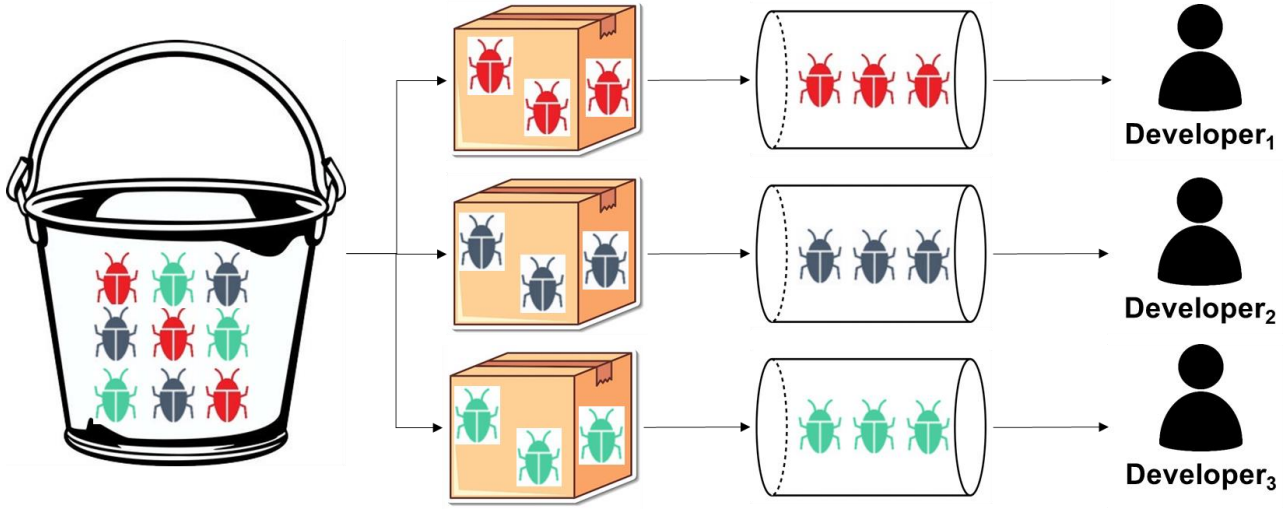
Using AI for Defect Management

How AI can help triage defects

- AI can support triage of identified defects
  - Defect categorization
  - Defect prioritization
  - Defect assignment



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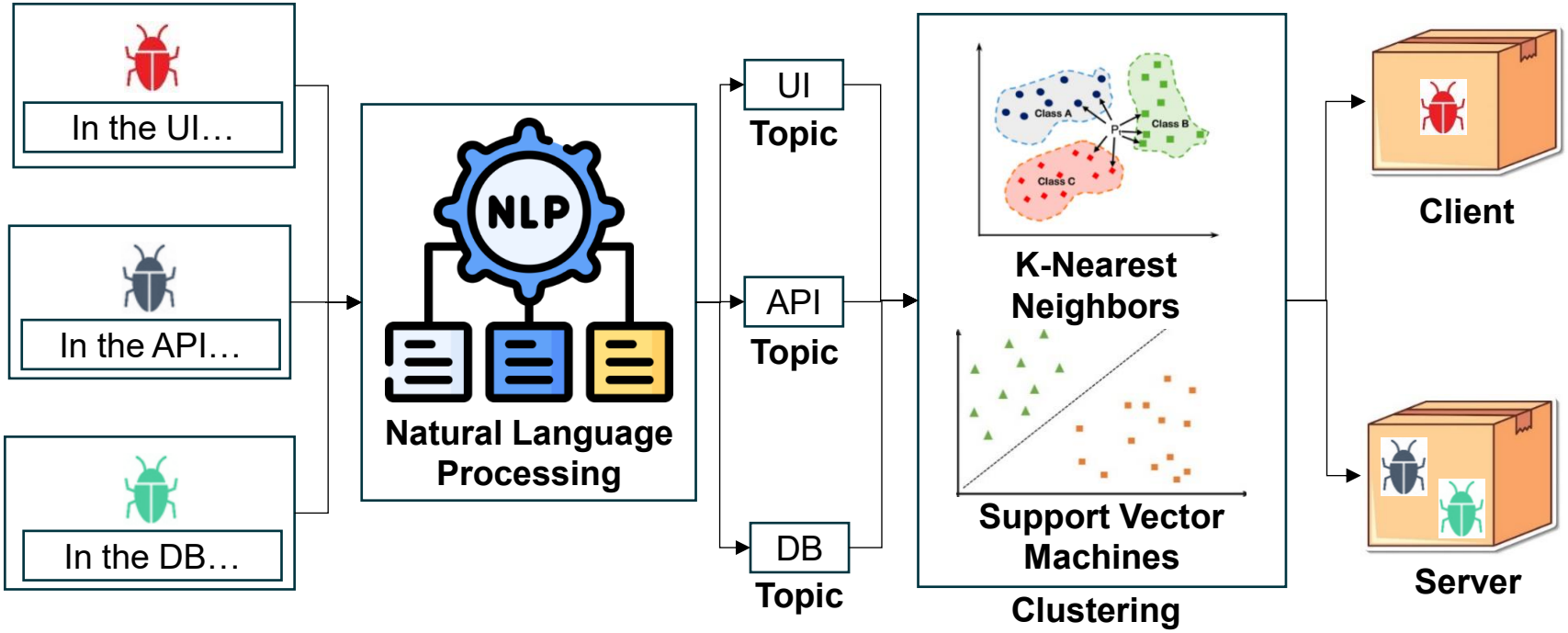
Using AI to Analyze Reported Defects

Categorizing Defects



Use natural language processing and clustering to categorize defects.

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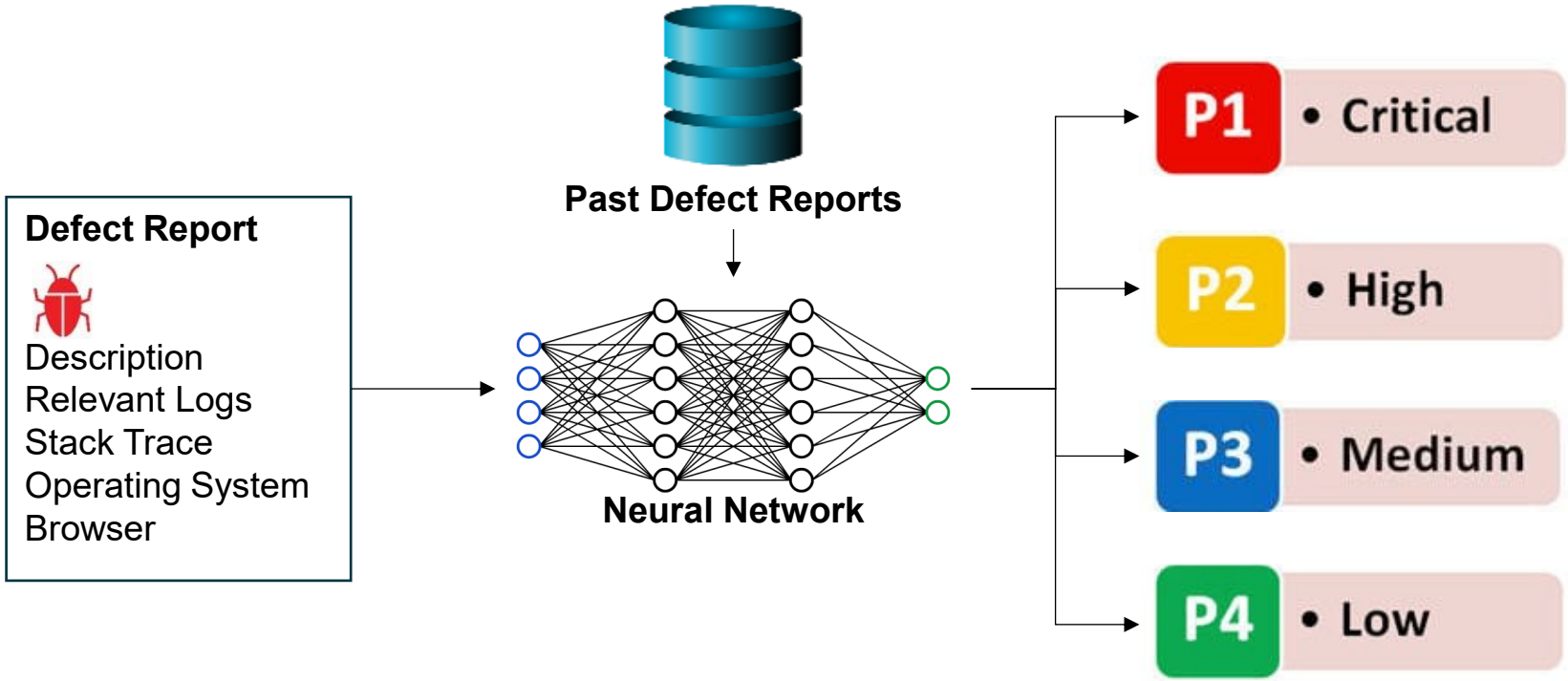
Using AI to Analyze Reported Defects

Evaluating Defect Priority



Utilize past defect reports to learn how new defects should be prioritized.

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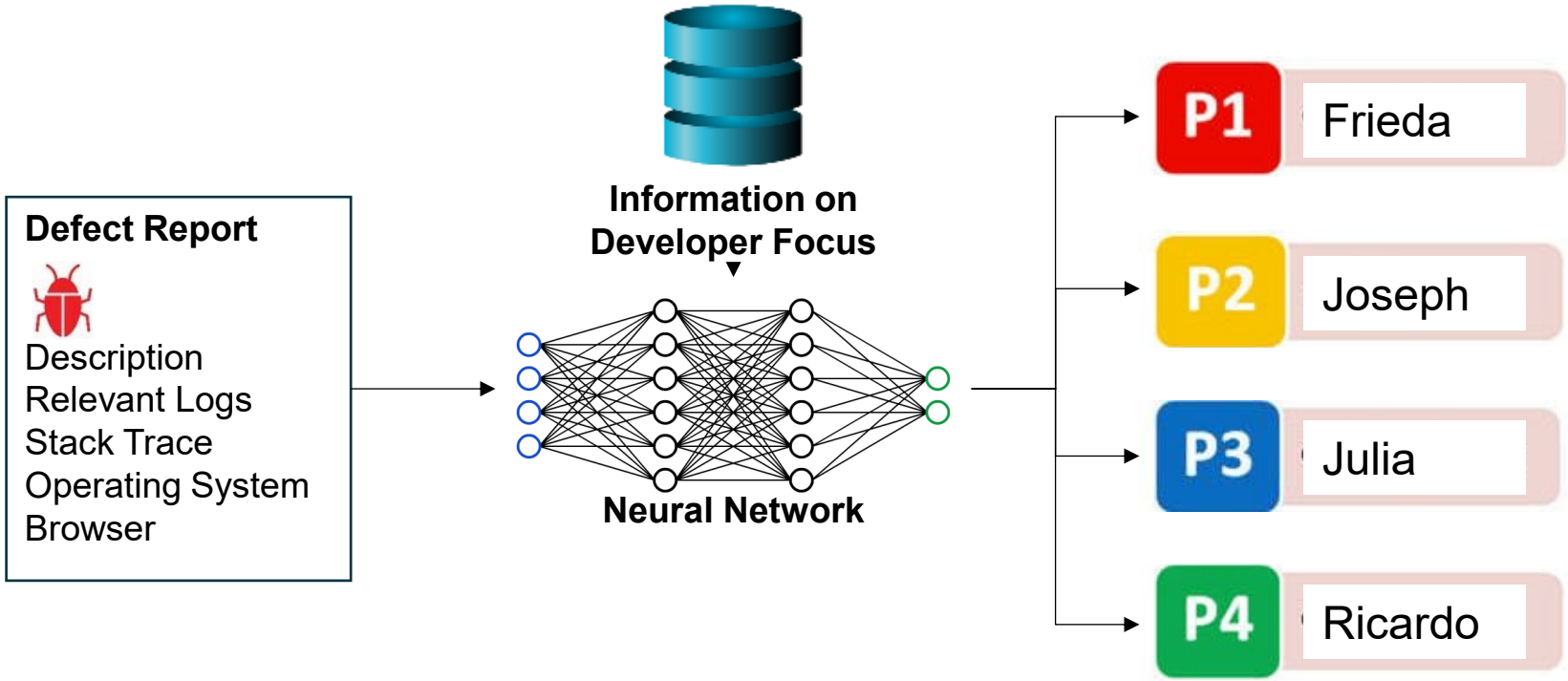
Using AI to Analyze Reported Defects

Assign Defects



Utilize information on developers to assign defects for remediation

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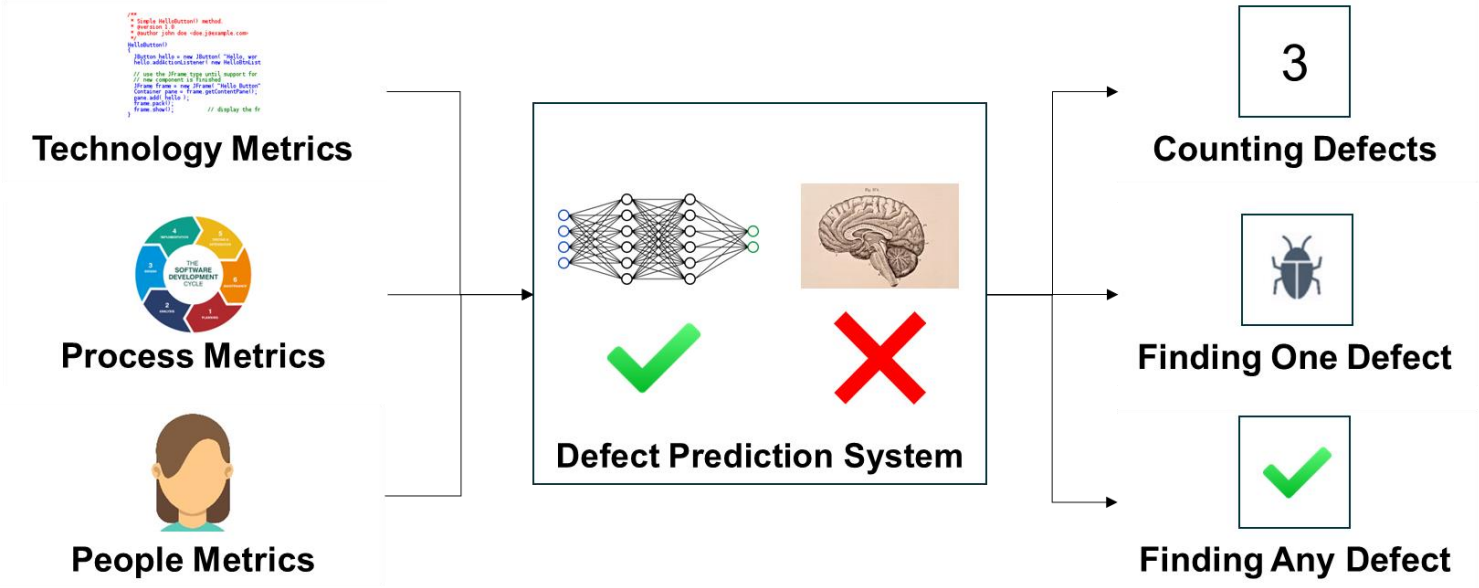
Using AI for Defect Prediction

Leverage information to predict faults

- AI can support defect prediction
  - Risk-based testing is part of this solution
  - Machine learning model based on technology, people, process metrics



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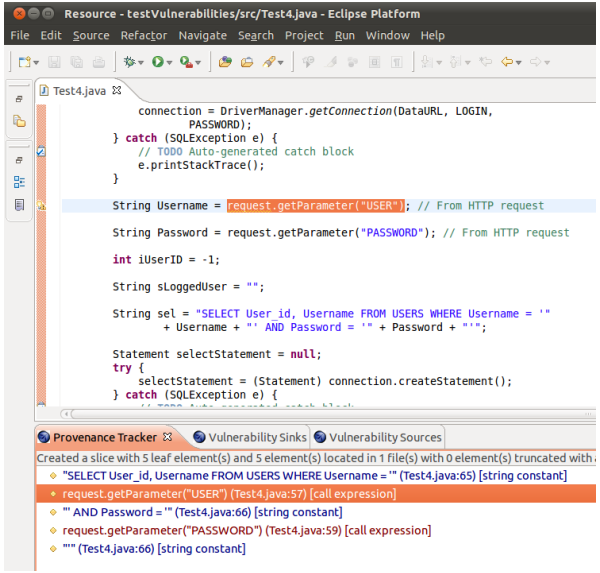
Using AI for Defect Prediction

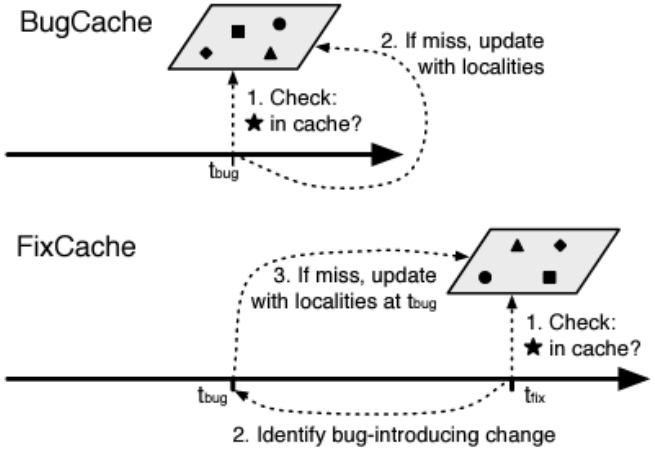
Research results

Technology Metrics



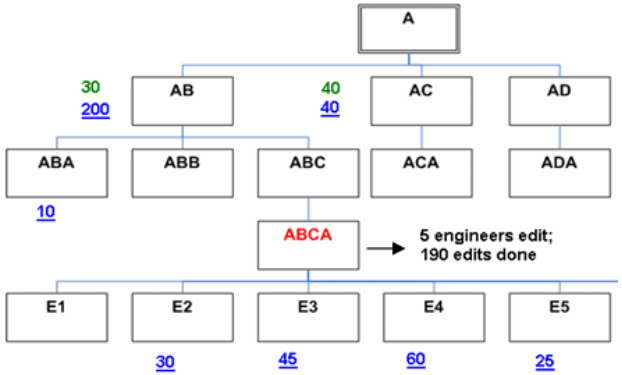
Accuracy: 88%

Process Metrics



Accuracy: 73-95%

People Metrics



Accuracy: 86.2%

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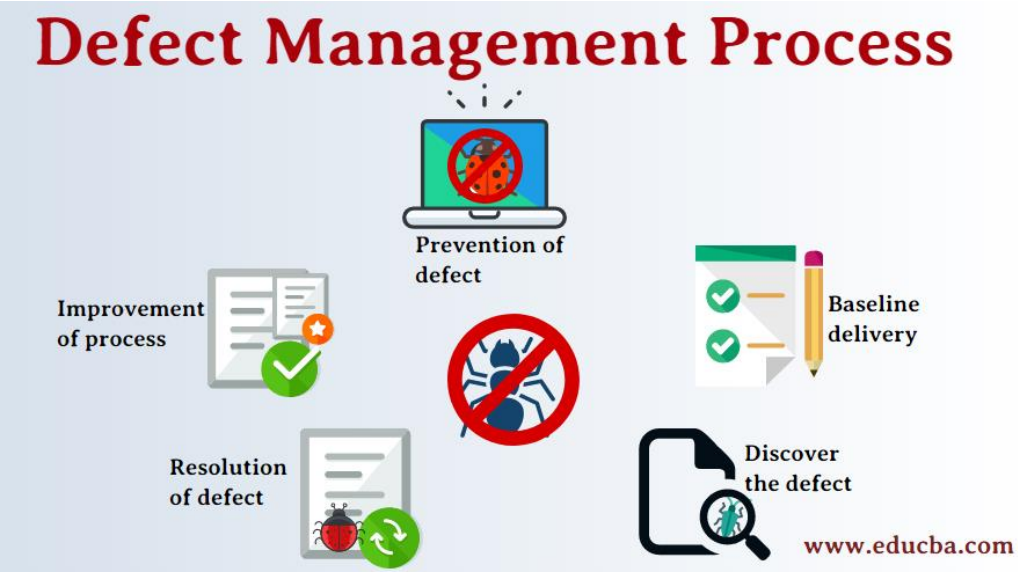
Workshop Demo and Exercise #6

Using AI to Manage Defects



- Follow along as the instructor shows how AI can manage defects
  - defect_management.ipynb
- Break into teams
- Create prompts to classify, prioritize, and assign the defects you found

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Optimizing Regression Testing

- AI can optimize your automated test suites
- Regression test suites grow as you iteratively build software and may become too large
- Optimization approaches
  - AI-based test case selection
  - AI-based test case prioritization
  - AI-based test combination/refactoring

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Workshop Demo

Using AI to Optimize Regression Testing

- What as the instructor shows how you might optimize your regression test suite based on its code coverage
  - test_suite_reduction.ipynb
- Discuss what other criteria could be used to optimize regression test suites

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Workshop Demo and Exercise #7

Using AI for Defect Prediction

- Watch as the instructor shows how AI can be used to create a defect prediction system
  - defect_prediction.ipynb
- Break into teams
- Try different criteria to see if you can increase accuracy

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7

Wrap Up

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Discussion

Share your Aha!

What questions do you still have?

How will you apply this in your current role?

What challenges do you foresee?

What are your next steps?

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Course Evaluation

How Did We Do?

We are committed to providing the highest quality education and certification products and to delivering a professional learning experience each and every time.

Your feedback is very important to us in evaluating course content, instructor delivery, learning methods and exercises, and more.

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Software Community

STICKYMINDS

A go-to resource for software professionals to stay current on the latest technologies and practices

www.stickyminds.com

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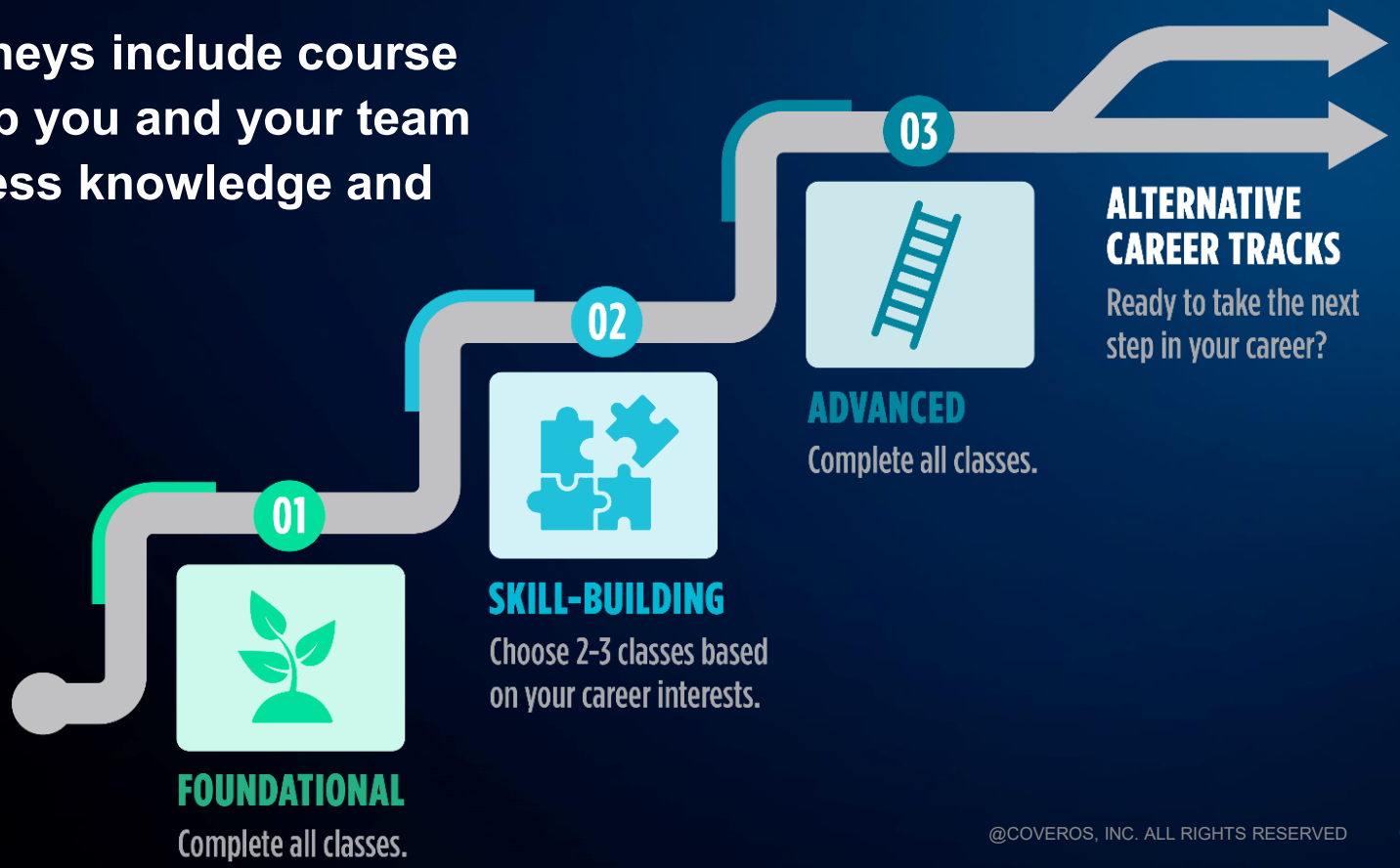
Next Steps

Continue Learning

Our Learning Journeys include course suggestions to help you and your team develop both process knowledge and technical skills.



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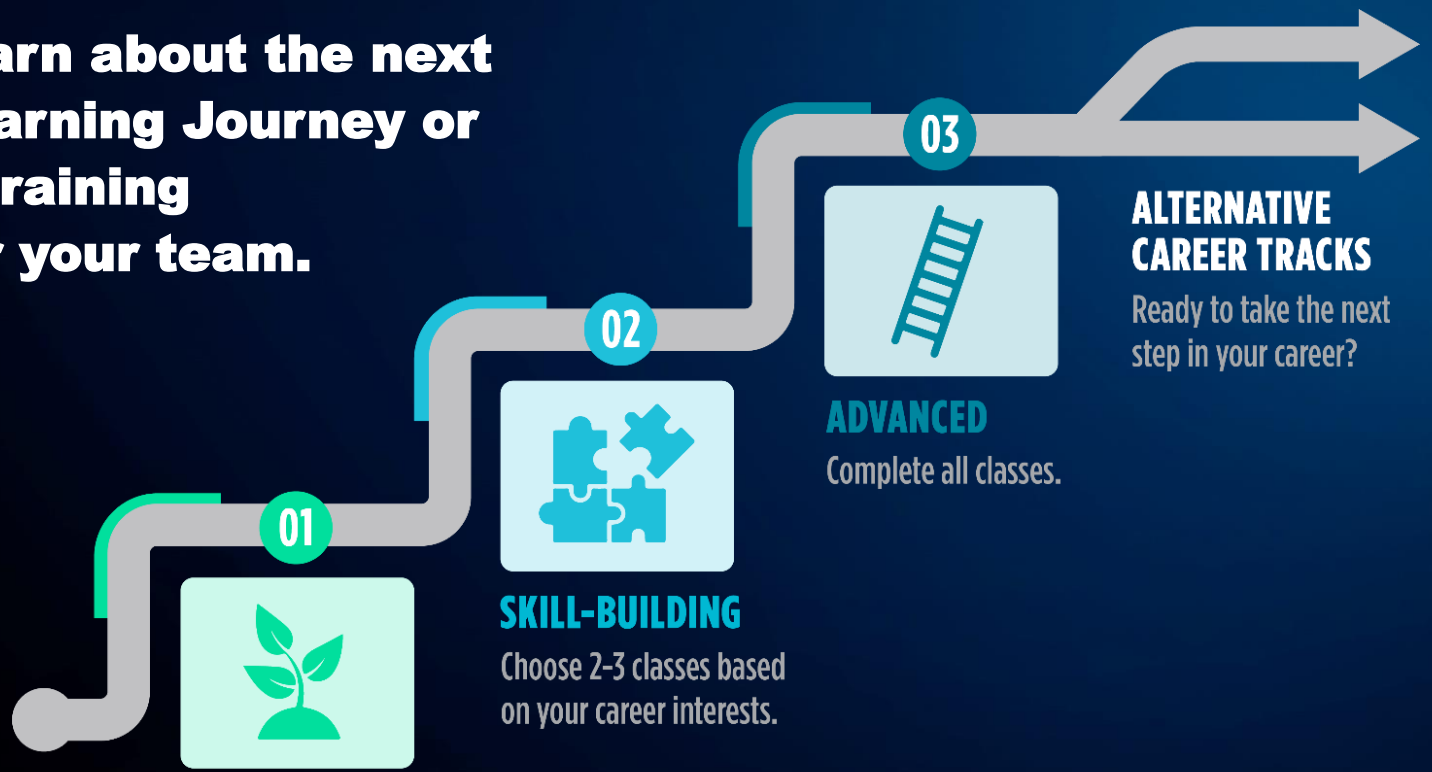
Next Steps

Continue Learning

Contact us to learn about the next stage of your Learning Journey or discuss private training opportunities for your team.



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Next Steps

Get Help with Key Initiatives

Does your team need help implementing new processes or delivering critical software? Ask us about how our experienced consultants can help with critical agile, DevOps, testing, or security initiatives.

Contact us to discuss consulting opportunities.

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